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D. The Seven Bridges of Königsberg

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given an **undirected unweighted connected** graph with N nodes and M edges. There can be self loop or multiple edges. Your task is to determine whether an Eulerian Path exists in the graph.

In graph theory, an Eulerian path (also called an Eulerian trail or Eulerian walk) is a path in a graph that visits every edge exactly once and may start and end at different vertices. However, a vertex can be visited multiple times.

Input

The first line contains two integers N and M ($1 \leq N \leq 2 \times 10^5, 0 \leq M \leq 3 \times 10^5$) — the number of vertices and the total number of edges.

The second line contains M integers $u_1, u_2, u_3 \dots u_m$ ($1 \leq u_i \leq N$) — where the i -th integer represents the node that is one endpoint of the i -th edge.

The third line contains M integers $v_1, v_2, v_3 \dots v_m$ ($1 \leq v_i \leq N$) — where the i -th integer represents the node that is other endpoint of the i -th edge.

The i 'th edge of this graph is between the i 'th node in the second line and the i 'th node in the third line.

Output

If an Eulerian Path exists, print YES. Otherwise, print NO.

Examples

input	Copy
5 10 5 5 5 2 2 2 3 3 4 2 2 3 1 3 4 1 4 1 2 4	
output	Copy
YES	
input	Copy
5 4 1 4 3 2 4 3 2 5	
output	Copy
YES	
input	Copy
8 7 4 4 6 6 3 1 8 6 5 3 2 7 8 7	
output	Copy
NO	
input	Copy

CSE221 Assignment 04 Summer 2026

Contest is running

17:30:05

Contestant



→ Submit?

Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



Submit

	7 6
	3 5 7 6 4 2
	5 7 6 4 2 1
output	
Copy	
YES	

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